

## Exercise sheet 3

Decisions – input, output, and programs that react

Not graded, not handed in. Do them anyway. This is where the learning happens! Solutions are published after we discuss them in class.

**Predict before you run.** Write the answer down first, then check it in Python. On the exam there will be no Python, so the paper answer is the one that counts.

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### Part A — Talking to the user

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#### A1 *Predict the f-string*

Given

```
n = 4
pi = 3.14159
name = "Ada"
```

write down exactly what each line prints. The | is there so you can see the spaces — count them.

```
1. print(f"{n} squared is {n**2}")
2. print(f"{pi:.2f}")
3. print(f"{pi:.0f}")
4. print(f"{n:>5}|")
5. print(f"{name:<5}|")
6. print(f"{name:^7}|")
7. print(f"{n/2}")
8. print(f"{1234567:,}")
```

Number 3 is not what most people expect. Number 7 is a reminder from last week.

#### A2 *They forgot int()*

Each program asks the user for a number and then gets it wrong. For each one: say whether it **raises an error** (name it) or **silently does the wrong thing**, and fix it.

|                                                                                              |                                                                                |
|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <pre># 1 age = input("Age: ") print("Next year:", age + 1)</pre>                             | <pre># 3 price = input("Price: ") print(f"With VAT: {price * 1.22:.2f}")</pre> |
| <pre># 2 a = input("First number: ") b = input("Second number: ") print("Sum:", a + b)</pre> | <pre># 4 n = input("How many? ") if n &gt; 10:     print("a lot")</pre>        |

One of these four is far more dangerous than the other three. Say which, and why.

### A3 Rewrite as f-strings

```
1. print("Total:", total, "euros")
2. print("Name: " + name + ", age: " + str(age))
3. print(mass, "/", height, "** 2 =", mass / height ** 2)
```

In number 3, also make the result show two decimal places.

### A4 Ask and answer

Write a program that:

- a) asks for a name;
- b) asks for a year of birth;
- c) prints Hello Ada, you turn 22 this year.

Use 2026 for the current year. Write the comment plan first.

Then answer in one sentence: which of the two inputs needed `int()`, and why did the other one not?

### A5 A table that stays lined up

Print this, with every number computed rather than typed:

|          |     |        |      |
|----------|-----|--------|------|
| Coffee   | 3 x | 1.50 = | 4.50 |
| Cornetto | 2 x | 1.15 = | 2.30 |
| Spremuta | 1 x | 3.00 = | 3.00 |
| Total    |     |        | 9.80 |

Now change Cornetto to Brioche integrale and re-run it. If your columns survived, you did it right. If they did not, you have spaces where you should have widths.

### A6 sep and end

What does this print? Write it out exactly, including the line breaks.

```
print("a", "b", "c", sep="-")
print(1, 2, sep="", end="")
print(3)
```

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## Part B — Making decisions

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### B1 True or False

Given `x = 5`, `y = 10` and `s = "Python"`:

- 1. `x < y` and `y < 20`
- 2. `x > y` or `y == 10`
- 3. `not (x == 5)`

4.  $0 < x < 10$
5.  $x == 5$  and  $y == 5$
6.  $s == \text{"python"}$

For number 6, say in a few words what you would write if you wanted it to be True.

## B2 Which branch runs?

For each program, say what it prints.

|                                                                                                                         |                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <pre># 1 temp = 30 if temp &gt; 30:     print("hot") elif temp &lt; 10:     print("cold") else:     print("fine")</pre> | <pre># 3 n = 0 if n:     print("something") else:     print("nothing")</pre>                          |
| <pre># 2 x = 7 if x &gt; 5:     print("big") if x &gt; 3:     print("medium")</pre>                                     | <pre># 4 mark = 30 if mark &gt;= 18:     print("pass") if mark &gt;= 28:     print("excellent")</pre> |

Number 2 prints two lines. Explain why, and say what you would change if you wanted only one.

## B3 Wrong, but no error

Each of these runs happily and gives a wrong answer for at least one input. Find the input that breaks it, explain what goes wrong, and fix it.

```
# 1 - a mark of 28 or 30 is a distinction
if mark == 28 or 30:
    print("distinction")

# 2 - 10% over 50 euros, 20% over 100
if amount > 50:
    discount = 0.10
elif amount > 100:
    discount = 0.20
else:
    discount = 0

# 3 - hot above 30, cold below 10, otherwise fine
if temp > 30:
    print("hot")
if temp < 10:
    print("cold")
if temp >= 10:
    print("fine")
```

These three are worth more than the rest of the sheet put together. A program that crashes tells you where to look. These do not.

**B4** *Ticket price*

A museum charges:

- free under 6;
- €5 from 6 to 17;
- €12 from 18 to 64;
- €8 from 65.

Write a program that asks for an age and prints the price. Write the comment plan first — one comment per band, *in the order you will test them*.

Then check your program on 5, 6, 17, 18, 64, 65 and 100. Two of those seven are the ones that might catch you.

**B5** *Two conditions*

Write a program that asks for a username and a password and prints `Welcome` only if the username is `ada` **and** the password is `lovelace`. Otherwise it prints `Access denied`.

- Write it with `and`.
- Write it again with two nested `if` statements.
- Which version would you rather read? Why?

*(Nobody checks passwords like this. We will see why in lecture 9.)*

**B6** *The discount, properly*

Write the discount program from **B3.2** correctly, as a complete program that asks for the amount and prints what there is to pay.

Test it on €40, €50, €50.01, €100 and €150.

**B7** *Air quality*

The CO<sub>2</sub> concentration in a lecture room, in parts per million, is a decent proxy for how stuffy it is:

- below 800: fine;
- 800 to 1400: getting stuffy;
- above 1400: open a window.

Write a program that asks for a reading and prints the right message.

- What does your program do for exactly 800? For exactly 1400?
- Is that what the three rules above say should happen?
- If not, fix either the program or the rules — and say which one you fixed.

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## If you finish early

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Go back to **B4** and add a second question: is it a weekday or the weekend? Tickets cost €2 more at the weekend, and the under-6s still go free.

You will find that one extra question doubles the number of branches. Keep the program; we will come back to it when we have the tool that fixes this.